Student Performance Analytics System

Course: Python with AI

Student Name : Bablu Hiralal Nishad

Institute: EWB Cource

Branch / Department: Computer Science & Engineering (CSE)

1. Project Title

Student Performance Analytics System

A Python-based data analytics application designed to automate student marks evaluation, performance calculation, grading, pass/fail identification, and subject-level statistical summary.

2. Objective

The objective of the Student Performance Analytics System is to analyze student academic performance using Python, NumPy, and Pandas. The project processes student marks, calculates subject-wise totals and averages, assigns grades, identifies pass or fail status, and summarizes overall class performance.

The system also identifies the highest-performing and lowest-performing students, calculates subject-wise class averages, and displays the top performers. This project demonstrates the use of Python functions, loops, conditional statements, NumPy numerical operations, and Pandas DataFrames for student performance analysis.

3. Technologies Used

- Python
- NumPy
- Pandas

4. Dataset Description

The dataset contains academic and attendance records for 15 students from the Computer Science & Engineering (CSE) branch.

5. Implementation Details

The project is implemented through the following structured steps:

1. **Dataset Creation and Storage:**
    - Raw data is structured as a Python dictionary containing student details and marks across 4 subjects (`python`, `java`, `dbms`, `dsa`) alongside attendance percentage.
    - It is converted into a Pandas DataFrame and exported to `student_data.csv`.
2. **Data Ingestion:**
    - The CSV file is read using `pd.read_csv('student_data.csv')` to ensure proper file handling and reload capability.
3. **Status & Grade Assignment Logic:**
    - **Passing Rule:** A student must score at least 40 marks in every individual subject. If marks in any subject are less than 40, the status is marked as `FAIL`; otherwise, `PASS`.
    - **Grading Rule:**
        - Total: sum of the three subject marks.
        - Average: total divided by 3.
        - A: average marks of 90 or above
        - B: average marks of 75 to below 90
        - C: average marks of 60 to below 75
        - D: average marks of 35 to below 60
        - F: average marks below 35
4. **Statistical Analysis:**
    - Overall pass and fail count calculation.
    - Overall class average score computation using `.mean()`.
    - Identification of highest and lowest performers using `.idxmax()` and `.idxmin()`.
    - Subject-wise statistical analysis generating the Mean, Maximum, and Minimum marks for each subject.

6. Key Features

- **CSV Data Persistence:** Automatically writes and reloads student records via CSV.
- **Subject-Level Condition Check:** Accurately marks students as `FAIL` if they fail any individual subject, regardless of their overall average.
- **Multi-tier Grading:** Classifies academic performance into 6 standardized grade tiers (A+, A, B, C, D, F).
- **Performance Outlier Identification:** Detects the topper and the lowest-performing student instantly.
- **Subject-Wise Insights:** Calculates Mean, Max, and Min marks for each subject to evaluate overall classroom difficulty.

7. Output/ Screenshot

- The Students Result
- The Student Performance

```
All Students Record:
    roll_no          s_name branch  total_marks  avg_marks grade status
0        20    Bablu Nishad    CSE          341      85.25     A   PASS
1        19    Rohit Sharma    CSE          370      92.50    A+   PASS
2        18      Amit Verma    CSE          285      71.25     B   PASS
3        17    Sachin Yadav    CSE          335      83.75     A   PASS
4        16    Rajesh Singh    CSE          185      46.25     F   PASS
5        15     Naman Gupta    CSE          375      93.75    A+   PASS
6        14   Aditya Bharti    CSE          245      61.25     C   PASS
7        13   Manish Thakre    CSE          284      71.00     B   PASS
8        12  Abhishek Yadav    CSE          364      91.00    A+   PASS
9        11     Vikas Yadav    CSE          145      36.25     F   FAIL
10       21     Rohan Donge    CSE          334      83.50     A   PASS
11       22    Manoj Sharma    CSE          235      58.75     D   PASS
12       23    Vivek Mishra    CSE          351      87.75     A   PASS
13       24    Suresh Singh    CSE          276      69.00     C   PASS
14       25    Nikhil Kumar    CSE          383      95.75    A+   PASS
```

```
        Final Summary
Total Students: 15
Passed: 14
Failed: 1
Class Average Marks: 75.13

Highest Scorer: Nikhil Kumar with 95.75
Lowest Scorer: Vikas Yadav with 36.25

Subject Averages:
PYTHON -> Mean: 75.73 | Max: 96 | Min: 35
JAVA -> Mean: 74.27 | Max: 95 | Min: 40
DBMS -> Mean: 75.73 | Max: 98 | Min: 38
DSA -> Mean: 74.8 | Max: 95 | Min: 32
```

8. Final Outcome

The program successfully executed of 15 student records without errors.

- Accurately categorized 14 passing students and 1 failing student (Vikas Yadav, who scored less then 40 in subjects like Python, DBMS, and DSA).

- Recognized **Nikhil Kumar** as the Highest Scorer 95.75 average marks and **Vikas Yadav** is the Lowest Scorer 36.25.
- Concluded an overall class average performance of **74.45**

9. Challenges & Learning

- **Challenge:** Handling multi-condition evaluation where a student might achieve a passing overall average but fail due to low scores in a specific subject.
    - *Solution:* Implemented conditional checks inside row iteration (`iterrows()`) to check all four subject thresholds before assigning status.
- **Learning:** Gained hands-on experience in using Pandas DataFrames, handling CSV file input/output, calculating column-level descriptive statistics, and formatting terminal output effectively.

10. GitHub Repository

- **Repository URL:** https://github.com/bablunishad-ui/student-performance-analytics